ECON3700 Environmental Economics

Semester 2, 2026

Individual Applied Case Study – WIL Industry Report

ALL SECTIONS DUE, 3pm 9th November. Submitted online.

Assignment Overview:

This assignment will consist of an Industry Integrated Work-Integrated Learning (WIL) report task in collaboration with Natural Capital & Climate worth **30%** (Written Report and Spreadsheet), and reflective writing worth **5%**, for a total of **35% of your final grade**.

The assignment must be submitted electronically through the Online Submission links in the Assessment section of the Course Blackboard site.

- **Part 1 Written Report** MUST be submitted as a Word file (.doc or docx)
- **Part 1 Spreadsheet Answers** MUST be submitted as an Excel file (.xls or .xlsx)
- **Part 2 Reflective Writing** MUST be submitted as a Word file (.doc or .docx)

Further details to be announced on Blackboard.

Written Report on Unsealed Road Management in Cook Shire

Objective:

Prepare a written report that evaluates alternative unsealed road management options in Cape York catchments using both Cost-Effectiveness Analysis (CEA) and Cost-Benefit Analysis (CBA) frameworks. The report should assess any economic, environmental, and social trade-offs of measures that reduce fine sediment delivery from road erosion.

Part 1 – Written Report Task – 100 marks (30%)

1. **Introduction:**
   - i. Provide a brief overview of the issues with existing road maintenance in the region and the resulting impacts from road erosion, focusing on the climate events in the region and risks to unsealed roads.
   - ii. Discuss the environmental, social, and economic implications of road erosion from the region (e.g., impacts to road access, fine sediment delivery to sensitive environments, ongoing cost to the broader society).
   - iii. Describe the project goals and summarise the four road management scenarios — the scenarios and their descriptions will be provided during the Industry Expert's presentation in **Week 12's Lecture**.
2. **Cost-Effectiveness Analysis (CEA):**
   - i. Each scenario reduces the volume of fine sediment (tonnes/year, <20 µm) eroded from the road surface, drains and batters, and delivered downstream. Compare the cost per unit of water quality improvement (i.e. cost per tonne of fine sediment avoided per year) across the scenarios using the data provided. These costs should include the lifecycle cost of each scenario over the 30-year assessment period.
   - ii. Analyse which scenario delivers the most cost-effective reduction in fine sediment delivery.
3. **Cost-Benefit Analysis (CBA):**
   - i. Identify and quantify the benefits of each scenario (e.g. avoided fine sediment pollution valued using an avoided abatement cost approach and reduced ongoing road maintenance costs). You may use the data provided and any additional research to quantify the benefits.

ii. Using the lifecycle costs, calculate net present values (NPV) for each scenario relative to the business-as-usual base case, applying the discount rate and assessment period specified in the spreadsheet.

iii. Discuss the challenges in quantifying non-market environmental benefits, such as improved reef water quality and habitat condition, the uncertainty in the avoided abatement cost value, and the risks of double counting.

4. **Comparison of CBA and CEA Results:**

i. Compare the outcomes from both CBA and CEA for each road management scenario.

ii. Discuss how different decision-makers can use these analyses to choose the most suitable road management approach and maximise value from public funding.

5. **Conclusion:**

Summarise your findings and provide a recommendation on the most economically feasible road management scenario. Highlight any trade-offs, uncertainties, or additional considerations that may affect long-term success, such as increasing rainfall intensity under climate change, the remoteness of works, and funding rules for disaster recovery.

Report Guidelines:

i. **Length**: A strict limit of 1,500 words.

ii. **Format**: Use an academic format (Times New Roman, 12pt, double-spaced, APA referencing style). *Refer to Week 13 lecture for more details on the report structure*.

iii. **Data Sources:** Spreadsheet provided. Project description and road management scenarios provided in *Lecture Week 12 – Industry Expert's presentation*.

Part 1 – Spreadsheet Answers

Use the provided spreadsheet of data to apply the CEA and CBA calculations. Save the file with your last name, first name, and student ID, and submit it through Blackboard.

General Assessment Criteria for Part 1:

i. Depth of Analysis – 40 marks

a. Ability to analyse environmental economics in real-world environmental issues.

b. Discussion of environmental, social and economic implications presented by industry expert.

c. Application of CBA and CEA methods.

d. Quality of recommendations based on economic and environmental outcomes.

ii. **Correctness – 40 marks (Including Excel submission)** *A mark of **zero** will be awarded if the spreadsheet is not submitted*.

   a. Accuracy of calculations using CBA and CEA methods.

   b. Effective use of tables and graphs to enhance clarity and understanding.

iii. Organisation and Structure – 10 marks

   a. Organisation and flow of report.

   b. Logical flow of arguments.

iv. Formatting – 10 marks

   a. Correct formatting and APA referencing and citations.

   b. Correct APA referencing and citations.

Refer to the rubric for more detailed criteria for Part 1. The spreadsheet answers are included as part of the rubric requirements.

Part 2 – Reflective Writing – 10 marks (5%)

In your reflective writing, use the 5R's framework (Reporting, Responding, Relating, Reasoning, and Reconstructing) to address the following prompt:
Reflect on your experience with the Industry Expert during their presentation. What insights did you gain about the application of CEA and CBA in real-world environmental projects? How did this experience shape your understanding of the challenges and opportunities in managing unsealed roads to protect downstream water quality?

Strict 400-word limit.

Refer to Week 13 lecture for more details on the 5R's framework and note that the rubric can be found on the course website.